

Sydney Casey

Data Scientist | Health Data Science MSc | Python ▪ SQL ▪ R ▪ ML ▪ Healthcare Analytics
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KEY COMPETENCIES

- **Programming languages:** Python, SQL, STATA, R
- **Data Science & Machine Learning:** Mathematical modeling, Model Evaluation, Hypothesis testing, Data science pipeline (cleansing, wrangling, visualization, modeling, interpretation), Python (eg. scikit-learn, numpy, pandas, matplotlib)
- **Data Visualization & Analysis Tools:** Tableau, Power BI, Matplotlib, Seaborn, JupyterLab, Streamlit, R Shiny
- **Additional Tools & Technologies:** Microsoft Office, Excel, MySQL, PostgreSQL, Docker, Survival Analysis, Statistics, Experimental Design, Apache Airflow, API integration, Microsoft Copilot, EPIC
- **Technical Reporting:** Strong skills in scientific reporting and presenting data to diverse audiences

EDUCATION

Master of Science in Health Data Science, University College London (UCL) Sept. 2022 - Sept. 2023
GPA: 3.75 / 4.00 (UK Distinction)

Relevant coursework: machine learning for healthcare & biomedicine, advance machine learning for healthcare, regression modeling, advanced statistics for records research, data methods for health research, programming with python for health research

Bachelor of Arts in Biochemistry, Mount Holyoke College May 2016
Cum laude, Mount Holyoke Leadership Award

PROFESSIONAL EXPERIENCE

Clinical Research Assistant June 2025 - Present
The Ohio State University Wexner Medical Center, Columbus, OH

- Analyze clinical trial datasets using R and Python to evaluate treatment outcomes and support data-driven research decisions
- Contribute to patient-facing AI chatbot project, collaborating with computer science and clinical teams to improve cardiovascular patient education
- Manage data collection and quality assurance across multiple vascular surgery clinical trials

Research Assistant Sept. 2020 - Aug. 2021
Nationwide Children's Hospital, Columbus, OH

- Performed statistical analysis to quantify experimental results including growth curves and protein expression levels.
- Presented weekly data analyses to research team, communicating findings that guided experimental design decisions.

Postbaccalaureate Award Fellow April 2017 - June 2019
National Institutes of Health, Bethesda, MD

- Conducted quantitative analysis of experimental data to assess disease mechanisms and treatment effects
- Delivered research presentations effectively communicating complex findings to diverse scientific audiences.

SELECTED PROJECTS

Predicting Length of Critical Care Days in Welsh Intensive Care | UCL June 2023 - Sept. 2023

- Built machine learning model predicting critical care duration from EHR data (MAE: 1.7 days, 89% sensitivity)
- Processed and cleaned 10,000+ patient records; engineered 30+ clinical features
- Identified top predictive factors using SHAP analysis to support clinical decision-making
- Validated model generalizability by applying to COVID-19 patient data (unseen during training), demonstrating robustness across different patient populations

Clinical Quality Analytics - Synthetic Patient Data | Interview Project Nov. 2024

- Analyzed multi-site clinical trial data evaluating demographics, protocol deviations, adverse events, and total exposure years
- Applied normal and Poisson distributions in R to establish quality tolerance limits and identify site-level anomalies
- Performed statistical process control analysis to detect site-level anomalies and flag high-risk patterns in patient safety data
- Developed data visualizations to communicate quality metrics and risk findings to stakeholders

Biomedical Literature Discovery Tool | Personal Project 2024

- Developed Python web application (Streamlit) enabling search and retrieval of articles from PubMed's 30M+ database via API integration
- Extracted and structured publication metadata (authors, dates, journals) into downloadable CSV format
- Created network graph visualizations using NetworkX and Plotly to map keyword co-occurrence patterns and research domain connections

PUBLICATIONS

3 peer-reviewed publications in biomedical research (neuronal ceroid lipofuscinoses, peritoneal disease) - available upon request